First Order ODEs

The **general form** of a first order ODE is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

for some given function f.

1 Linear

The **general form** of a first order linear ODE is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

for some given functions p and q.

1.1 Method of Integrating Factors

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

the **general solution** is

$$y = \frac{1}{\mu(t)} \int \mu(t) \cdot q(t) \, \mathrm{d}t$$

where the **integrating factor** is any $\mu(t)$ that satisfies

$$\ln |\mu(t)| = \int p(t) \, \mathrm{d}t$$

Remark. Indefinite integrals yield a family of functions differing by a constant. Each indefinite integral results in an independent constant, which represents a degree of freedom. However, the constant from $\mu(t)$ cancels out in the final solution for y.

1.2 Variation of Parameters

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

let $h(t)$ be a function such that

$$h(t) = \int p(t) \, \mathrm{d}t$$

Then the **general solution** is given by

$$y = e^{-h(t)} \int q(t) \cdot e^{h(t)} \, \mathrm{d}t$$

1.3

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

the **general solution** is given by

$$y = v \int \frac{q(t)}{v} \, \mathrm{d}t$$

where v satisfies

$$\frac{\mathrm{d}v}{\mathrm{d}t} + p(t) \cdot v = 0$$

Proof.

We want to solve eq. (1). Suppose v satisfies eq. (2), which is separable and can be solved easily.

$$y' + p(t) \cdot y = q(t) \tag{1}$$

$$v' + p(t) \cdot v = 0 \tag{2}$$

Take the derivative of y/v:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{y}{v}\right) &= \frac{y'v - v'y}{v^2} \\
&= \frac{y'v + (p(t) \cdot v)y}{v^2} && \text{Plug in eq. (2).} \\
&= \frac{q(t)}{v} && \text{Plug in eq. (1).}
\end{aligned}$$

Thus we have

$$\frac{y}{v} = \int \frac{q(t)}{v} \,\mathrm{d}t$$

Since v can be solved from eq. (2), we obtain the solution for y. □

2 Non-Linear

2.1 Separable Equations

A first order ODE is **separable** if it can be written in the form

$$M(t) + N(y) \cdot \frac{\mathrm{d}y}{\mathrm{d}t} = 0$$

for some functions M and N. Then the solution satisfies

$$\int N(y) \,\mathrm{d}y = -\int M(t) \,\mathrm{d}t$$

2.2 Transformation Methods

Some complicated non-linear ODEs can be transformed into ODEs that are solvable by suitable substitutions.

2.2.1 Bernoulli Equations

Bernoulli equations can be transformed into linear ODEs by the substitution $v = y^{1-r}$.

Definition 2.1 (Bernoulli Equations)**.**

A first order ODE is called a **Bernoulli equation** if it can be written in the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t) \cdot y^r \tag{3}$$

for some given functions p, q and a real number $r \in \mathbb{R} \setminus \{0, 1\}$.

The **general solution** of a Bernoulli equation (3) is given by

$$y^{1-r} = \frac{1-r}{\mu(t)} \int \mu(t) \cdot q(t) \,\mathrm{d}t$$

where $\mu(t)$ satisfies

$$\ln|\mu(t)| = (1-r) \int p(t) \,\mathrm{d}t$$

Note: If $r > 0$, then $y = 0$ is also a solution, which is not captured by the above formula.

Proof.

Substitute $v = y^{1-r}$, and then solve with the method of integrating factors.

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)\cdot y = q(t)\cdot y^r \quad &\Longrightarrow \quad y^{-r}\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)\cdot y^{1-r} = q(t) \quad \text{for} \quad y \neq 0 \\ &\Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}\left(y^{1-r}\right) + (1-r)p(t)\cdot y^{1-r} = (1-r)q(t)\end{aligned}$$

□

2.2.2 Homogeneous Equations

Homogeneous equations can be transformed into separable equations.

If a first-order ODE can be expressed in the following form, then it is said to be **homogeneous**:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = F(y/t)$$

for some given function F.

Then it can be transformed into a separable equation using $v = y/t$:

$$\frac{1}{F(v) - v}\,\mathrm{d}v = \frac{1}{t}\,\mathrm{d}t$$

Remark. The homogeneous equations considered here have nothing to do with the homogeneous LCCDEs.

2.3 Exact Equations

Definition 2.2 (Exact Equations)**.**

A first order ODE $M(t,y)\,\mathrm{d}t + N(t,y)\,\mathrm{d}y = 0$ is said to be **exact** if there exists a function $\Psi(t,y)$ such that

$$\frac{\partial\Psi(t,y)}{\partial t} = M(t,y) \quad \text{and} \quad \frac{\partial\Psi(t,y)}{\partial y} = N(t,y)$$

Then the general solution is given implicitly as

$$\Psi(t,y) = C \qquad \text{for some constant } C \in \mathbb{R}$$

Theorem 2.1.

Let R be a simply connected region in the ty-plane. If $M(t,y)$, $N(t,y)$, $M_y(t,y)$, and $N_t(t,y)$ are continuous in R, then

$$M(t,y) + N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \text{ is exact in } R \qquad \Longleftrightarrow \qquad M_y(t,y) = N_t(t,y) \quad \text{for all } (t,y) \in R$$

where $M_y(t,y) := \partial M(t,y)/\partial y$ and $N_t(t,y) := \partial N(t,y)/\partial t$.

To solve the first order ODE $M(t,y)\,\mathrm{d}t + N(t,y)\,\mathrm{d}y = 0$, we can follow these steps:

1. Verify that the equation is exact by checking if $M_y(t,y) = N_t(t,y)$.
2. Integrate $M(t,y)$ with respect to t to find $\Psi(t,y)$:

   $$\Psi(t,y) = \int M(t,y)\,\mathrm{d}t + h(y)$$

   where $h(y)$ is an arbitrary function of y.
3. Differentiate $\Psi(t,y)$ with respect to y and set it equal to $N(t,y)$:

   $$\frac{\partial\Psi(t,y)}{\partial y} = N(t,y)$$

This will allow us to solve for $h(y)$.

4. Substitute $h(y)$ back into $\Psi(t, y)$ to obtain the implicit solution:

$$\Psi(t, y) = C$$

for some constant C.

2.4 Exact Equations and Integrating Factors

It is sometimes possible to convert an ODE that is not exact into an exact equation with a suitable integrating factor. Suppose the following first order ODE is exact:

$$\mu M(t, y) + \mu N(t, y) \frac{\mathrm{d}y}{\mathrm{d}t} = 0$$

Then the integrating factor μ must satisfy

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial t}$$

On one hand, assuming that $\mu = \mu(t)$, we have

$$\frac{\mathrm{d}\mu}{\mathrm{d}t} = \frac{M_y - N_t}{N} \cdot \mu$$

If $(M_y - N_t)/N$ depends on t alone, then we can solve for $\mu(t)$.

On the other hand, assuming that $\mu = \mu(y)$, we have

$$\frac{\mathrm{d}\mu}{\mathrm{d}y} = \frac{N_t - M_y}{M} \cdot \mu$$

If $(N_t - M_y)/M$ depends on y alone, then we can solve for $\mu(y)$.

If neither case applies, then try some substitutions to transform the ODE into a solvable form.

3 The Existence and Uniqueness Theorem

3.1 The Theorem

Theorem 3.1 (Existence and Uniqueness Theorem for First Order Linear ODEs)**.**

For the IVP:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = p(t) \cdot y + q(t) \quad \text{and} \quad y(t_0) = y_0$$

if there exists an interval $I \subseteq \mathbb{R}$ such that $t_0 \in I$ and $p(t)$, $q(t)$ are **continuous** on I, then there exists a **unique** solution $y(t)$ defined on I that satisfies the IVP.

Remark. The solution is defined on the entire interval I.

Theorem 3.2 (Existence and Uniqueness Theorem)**.**

For the IVP:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y) \quad \text{and} \quad y(t_0) = y_0$$

if there exists a closed rectangle $R : t \in [t_0 - a, t_0 + a], y \in [y_0 - b, y_0 + b]$ such that $f(t, y)$ and $\partial f(t, y)/\partial y$ are **continuous** on R, then there exists a **unique** solution $y = \phi(t)$ defined for $t \in [t_0 - h, t_0 + h]$, where h is some number satisfying $0 < \min\{b/M, a\} \leqslant h \leqslant a$ and $M = \max\limits_{(t,y)\in R} |f(t, y)| < +\infty$.

Remark. The solution is guaranteed to be defined at least on the interval of t such that $|t - t_0| \leqslant \min\{b/M, a\}$, where M is the bound of $f(t, y)$ on the closed rectangle R.

Remark. If the conditions of the theorem are not satisfied, then the solution may not be unique or may not exist.

Theorem 3.3 (Existence Theorem)**.**

For the IVP:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t,y) \quad \text{and} \quad y(t_0) = y_0$$

if there exists a closed rectangle $R : t \in [t_0 - a, t_0 + a], y \in [y_0 - b, y_0 + b]$ such that $f(t,y)$ is **continuous** on R, then there exists **at least one** solution $y = \phi(t)$ defined for $t \in [t_0 - h, t_0 + h]$, where h is some number satisfying $0 < \min\{b/M, a\} \leqslant h \leqslant a$ and $M = \max\limits_{(t,y)\in R} |f(t,y)| < +\infty$.

Remark. When $\partial f/\partial y$ may not be continuous, the solution may not be unique.

3.2 The Preliminary before the Proof

Definition 3.1 (Pointwise Convergence)**.**

A sequence of functions $\{f_n\}$ defined on a set $\mathcal{D} \subseteq \mathbb{R}$ is said to **converge pointwise** to a function f if:

$$\forall x \in \mathcal{D}, \lim_{n\to\infty} f_n(x) = f(x)$$

The **rate of convergence** can be **different** for different points in $\mathcal{D}$.

Definition 3.2 (Uniform Convergence)**.**

A sequence of functions $\{f_n\}$ defined on a set $\mathcal{D} \subseteq \mathbb{R}$ is said to **converge uniformly** to a function f if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n > N, \forall x \in \mathcal{D}, |f_n(x) - f(x)| < \varepsilon$$

The **rate of convergence** is the **same** for all points in $\mathcal{D}$.

Uniform convergence implies pointwise convergence.

Proof.

The proof is given in first-order formal logic. The inference rules used are the following:

- $\alpha \to \beta \iff \neg\alpha \vee \beta$
- $\forall x(\alpha \vee P(x)) \iff \alpha \vee \forall x P(x)$ and $\exists x(\alpha \vee P(x)) \iff \alpha \vee \exists x P(x)$
- $\forall x \forall y P(x,y) \iff \forall y \forall x P(x,y)$ and $\exists x \forall y P(x,y) \implies \forall y \exists x P(x,y)$

To make it easier, we suppose all the variables are restricted to real numbers. Rewrite the definition of uniform convergence into first-order formal logic:

$$\begin{aligned}
\text{Uniform Convergence} &\iff \forall\varepsilon\Big(\varepsilon > 0 \to \exists N\Big[N \in \mathbb{N} \to \forall n\Big(n > N \to \forall x\big(x \in \mathcal{D} \to |f_n(x) - f(x)| < \varepsilon\big)\Big)\Big]\Big) \\
&\iff \forall\varepsilon\Big(\varepsilon \leqslant 0 \ \vee\ \exists N\Big[N \notin \mathbb{N} \ \vee\ \forall n\Big(n \leqslant N \ \vee\ \forall x\big(x \notin \mathcal{D} \ \vee\ |f_n(x) - f(x)| < \varepsilon\big)\Big)\Big]\Big) \\
&\implies \forall x\forall\varepsilon\Big(\varepsilon \leqslant 0 \ \vee\ \exists N\Big[N \notin \mathbb{N} \ \vee\ \forall n\Big(n \leqslant N \ \vee\ x \notin \mathcal{D} \ \vee\ |f_n(x) - f(x)| < \varepsilon\Big)\Big]\Big) \\
&\iff \forall x\Big[x \notin \mathcal{D} \ \vee\ \forall\varepsilon\Big(\varepsilon \leqslant 0 \ \vee\ \exists N\Big[N \notin \mathbb{N} \ \vee\ \forall n\Big(n \leqslant N \ \vee\ |f_n(x) - f(x)| < \varepsilon\Big)\Big]\Big)\Big] \\
&\iff \forall x\Big[x \in \mathcal{D} \to \forall\varepsilon\Big(\varepsilon > 0 \to \exists N\Big[N \in \mathbb{N} \to \forall n\Big(n > N \to |f_n(x) - f(x)| < \varepsilon\Big)\Big]\Big)\Big] \\
&\iff \forall x \in \mathcal{D}, \forall\varepsilon > 0, \exists N \in \mathbb{N}, \forall n > N, |f_n(x) - f(x)| < \varepsilon \\
&\iff \text{Pointwise Convergence}
\end{aligned}$$

$\square$

Theorem 3.4 (Weierstrass M-test)**.**

Let $\{\phi_n\}$ be a sequence of functions defined on a set $\mathcal{D} \subseteq \mathbb{R}$. If there exists a sequence of positive numbers $\{M_n\}$ such that

$$\forall n \in \mathbb{N}, \forall x \in \mathcal{D}, |\phi_n(x)| < M_n \quad \text{and} \quad \sum_{n=1}^{\infty} M_n \text{ converges}$$

then the new sequence $\{\Phi_n(x)\}$, where $\Phi_n(x) := \sum_{k=1}^{n} \phi_k(x)$, **converges uniformly** on $\mathcal{D}$.

Theorem 3.5.

Suppose $\{\phi_n(x)\}$ is a sequence of functions defined on an interval $[a, b] \subseteq \mathbb{R}$ and is **uniformly convergent**, then the operations of limit and integration can be interchanged:

$$\lim_{n\to\infty} \int_a^b \phi_n(x)\,\mathrm{d}x = \int_a^b \lim_{n\to\infty} \phi_n(x)\,\mathrm{d}x$$

3.3 The Proof

In this section, we will use the **method of successive approximations**, aka **Picard's iteration method**, to prove Theorem 3.2. Suppose the IVP is given by $\mathrm{d}y/\mathrm{d}t = f(t, y)$ and $y(t_0) = y_0$. And both the functions f and $\partial f/\partial y$ are continuous on the rectangle $R : t \in [t_0 - a, t_0 + a], y \in [y_0 - b, y_0 + b]$.

We start by defining a sequence of functions $\{\phi_n\}$ as follows:

$$\phi_0(t) = y_0$$
$$\phi_{n+1}(t) = y_0 + \int_{t_0}^{t} f(\tau, \phi_n(\tau))\,\mathrm{d}\tau \qquad \text{for } n = 0, 1, 2, \ldots$$

Note that each ϕ_n satisfies the initial condition $\phi_n(t_0) = y_0$. Then we will take the following steps to complete the proof:

1. Find the interval of t on which all ϕ_n are well-defined and continuous.
2. Show that $\{\phi_n(t)\}$ is uniformly convergent on that interval.
3. Show that the limit function $\phi(t) := \lim_{n\to\infty} \phi_n(t)$ is a solution to the IVP.
4. Show that the solution is unique.

Step 1. The Solution Interval

Since $f(t, y)$ is continuous on the rectangle $R : [t_0 - a, t_0 + a] \times [y_0 - b, y_0 + b]$, we need to find an interval of t such that $(t, \phi_n(t))$ is in R for all n, so that all $\phi_n(t)$ are well-defined and continuous on that interval. The result is $[t_0 - h, t_0 + h]$, where $h := \min\{b/M, a\}$ and $M := \max_{(t,y)\in R} |f(t, y)|$. We can verify this by mathematical induction.

First, it is clear that $\phi_0(t) = y_0$ is continuous and is restricted to R when $t \in [t_0 - h, t_0 + h]$. Now suppose the conclusion holds for $\phi_k(t)$. Then we have

$$\begin{aligned} |\phi_{k+1}(t) - y_0| &= \left| y_0 + \int_{t_0}^{t} f(\tau, \phi_k(\tau))\,\mathrm{d}\tau - y_0 \right| \\ &= \left| \int_{t_0}^{t} f(\tau, \phi_k(\tau))\,\mathrm{d}\tau \right| \\ &\leqslant \int_{t_0}^{t} |f(\tau, \phi_k(\tau))|\,\mathrm{d}\tau \end{aligned}$$

Since $f(t,y)$ is continuous on R and $(\tau, \phi_k(\tau))$ is in R for all τ in the integral, $f(\tau, \phi_k(\tau))$ is bounded, thus

$$
\begin{aligned}
|\phi_{k+1}(t) - y_0| &\leqslant \int_{t_0}^{t} |f(\tau, \phi_k(\tau))| \,\mathrm{d}\tau \\
&\leqslant \int_{t_0}^{t} M \,\mathrm{d}\tau \quad \text{where} \quad M := \max_{(t,y)\in R} |f(t,y)| \\
&= M|t - t_0| \leqslant Mh \leqslant b
\end{aligned}
$$

which implies that $\phi_{k+1}(t) \in [y_0 - b, y_0 + b]$ for all $t \in [t_0 - h, t_0 + h]$. And since $f(\tau, \phi_k(\tau))$ is continuous, the integral also results in a continuous function. Therefore, by induction, we have proven that the sequence $\{\phi_n(t)\}$ is well-defined and continuous on $[t_0 - h, t_0 + h]$.

Step 2. Uniform Convergence

We show that the sequence $\{\phi_n(t)\}$ is uniformly convergent using Theorem 3.4, the Weierstrass M-test.

Define $g_n(t) := \phi_n(t) - \phi_{n-1}(t)$ for $n = 1, 2, \ldots$, so that $\phi_n(t) = \phi_0(t) + \sum_{k=1}^{n} g_k(t)$. Then we need to find a sequence of positive numbers $\{M_n\}$ such that

$$
\forall t \in [t_0 - h, t_0 + h], |g_n(t)| < M_n \quad \text{and} \quad \sum_{n=1}^{\infty} M_n \text{ converges.}
$$

Since $\dfrac{\partial f}{\partial y}$ is continuous on R, for any $(t, y_1), (t, y_2) \in R$, there exists y_3 between y_1 and y_2 such that

$$
\frac{f(t,y_1) - f(t,y_2)}{y_1 - y_2} = \frac{\partial f(t,y)}{\partial y}\bigg|_{y=y_3}
$$

Since $\dfrac{\partial f}{\partial y}$ is also bounded on R, there exists $K > 0$ such that

$$
|f(t,y_1) - f(t,y_2)| \leqslant K|y_1 - y_2|
$$

Thus we have

$$
\begin{aligned}
|g_{n+1}(t)| &= |\phi_{n+1}(t) - \phi_n(t)| \\
&\leqslant \int_{t_0}^{t} \Big| f(\tau, \phi_n(\tau)) - f(\tau, \phi_{n-1}(\tau)) \Big| \,\mathrm{d}\tau \\
&\leqslant K \int_{t_0}^{t} |g_n(\tau)| \,\mathrm{d}\tau
\end{aligned}
$$

Let $M := \max_{(t,y)\in R} |f(t,y)|$.

$$
\begin{aligned}
|g_1(t)| &\leqslant \int_{t_0}^{t} |f(\tau, y_0)| \,\mathrm{d}\tau \\
&\leqslant M|t - t_0|
\end{aligned}
$$

By mathematical induction, we can prove that

$$
|g_n(t)| \leqslant \frac{MK^{n-1}|t - t_0|^n}{n!}
$$

Then, let $M_n := \dfrac{MK^{n-1}h^n}{n!}$, we have

$$
\begin{aligned}
|g_n(t)| &\leqslant M_n \qquad \forall t \in [t_0 - h, t_0 + h] \\
\sum_{n=1}^{\infty} M_n &= \frac{M}{K} \sum_{n=1}^{\infty} \frac{(Kh)^n}{n!} \\
&= \frac{M}{K}(e^{Kh} - 1) \quad \text{converges}
\end{aligned}
$$

Therefore, by Theorem 3.4, the sequence $\{\phi_n(t)\}$ is uniformly convergent on $[t_0 - h, t_0 + h]$.

Step 3. The Existence of the Solution

Let $\phi(t) := \lim_{n\to\infty} \phi_n(t)$. Then

$$\begin{aligned}
&\phi_{n+1}(t) = y_0 + \int_{t_0}^{t} f(\tau, \phi_n(\tau))\,\mathrm{d}\tau \\
&\implies \lim_{n\to\infty} \phi_{n+1}(t) = y_0 + \lim_{n\to\infty} \int_{t_0}^{t} f(\tau, \phi_n(\tau))\,\mathrm{d}\tau \\
&\implies \lim_{n\to\infty} \phi_{n+1}(t) = y_0 + \int_{t_0}^{t} \lim_{n\to\infty} f(\tau, \phi_n(\tau))\,\mathrm{d}\tau \qquad \text{by Theorem 3.5} \\
&\implies \phi(t) = y_0 + \int_{t_0}^{t} f(\tau, \phi(\tau))\,\mathrm{d}\tau \qquad \text{as } f(t,y) \text{ is continuous}
\end{aligned}$$

Therefore, $\phi(t)$ is a solution to the IVP.

Step 4. The Uniqueness of the Solution

Suppose $\psi(t)$ is another solution to the IVP. Then we have

$$\begin{aligned}
|\phi(t) - \psi(t)| &= \left| \int_{t_0}^{t} f(\tau, \phi(\tau)) - f(\tau, \psi(\tau))\,\mathrm{d}\tau \right| \\
&\leqslant K \int_{t_0}^{t} |\phi(\tau) - \psi(\tau)|\,\mathrm{d}\tau \quad \text{since } \frac{\partial f}{\partial y} \text{ is continuous and bounded on } R
\end{aligned}$$

Let $\Psi(t) := |\phi(t) - \psi(t)|$.

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\Psi(t) \leqslant K\Psi(t) \quad \text{and} \quad \Psi(t_0) = 0 \\
&\implies e^{-Kt}\left(\frac{\mathrm{d}}{\mathrm{d}t}\Psi(t) - K\Psi(t)\right) \leqslant 0 \\
&\implies \frac{\mathrm{d}}{\mathrm{d}t}\big(e^{-Kt}\Psi(t)\big) \leqslant 0 \\
&\implies \int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}\tau}\big(e^{-K\tau}\Psi(\tau)\big)\,\mathrm{d}\tau \leqslant 0 \\
&\implies e^{-Kt}\Psi(t) \leqslant e^{-Kt_0}\Psi(t_0) = 0 \\
&\implies \Psi(t) = 0 \qquad \forall t \in [t_0 - h, t_0 + h] \\
&\implies \phi(t) = \psi(t) \qquad \forall t \in [t_0 - h, t_0 + h]
\end{aligned}$$

Therefore, the solution to the IVP is unique.